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Hiring Committee
University of Southampton
Southampton, United Kingdom

To the Hiring Committee,

I am writing to express my strong interest in the AI-driven silicon photonics circuits design position at University of Southampton. As an AI & Data Science Engineer with a deep foundation in advanced neural network architectures, high-performance data pipelines, and mathematical modeling, I am highly motivated to contribute to your research goals.

In my recent work on the “Cardiovascular Medical Digital Twin,” I developed the **SST_KODA_MultiScale** architecture and **Medical_MambaTS** to process high-frequency 100Hz clinical telemetry data. A critical component of dealing with real-world medical data is handling uncertainty; to address this, I engineered a custom Missingness-Aware Gate to handle sensor dropouts dynamically.

I am eager to apply my PyTorch expertise and my passion for cutting-edge AI research to advance the initiatives at University of Southampton. Enclosed are my CV, university transcripts, and a detailed description of my technical projects.

Sincerely,

Ismail Aissaoui